



ya diggg: Physical Design & Material Specification

This document defines the mechanical technicalities for the ya diggg, ensuring the device meets the "Discreet, Premium, and Durable" design DNA [Medium](#).

1. Enclosure: Translucent "Soul" Shell

The outer shell utilizes a high-performance **Translucent Polycarbonate (PC)** to achieve the 90s analog aesthetic while maintaining durability.

Spec	Value	Technical Rationale
Material	PC (Smoke Amber Tint)	High impact resistance, light transmission tuned to 35% for "discreet" component visibility. Polycarbonate-Panels .
Wall Thickness	1.5mm - 1.8mm	Optimal balance between structural rigidity and weight. Bolt fasteners .
Finish	Frosted / Matte	Fingerprint resistance and sophisticated light diffusion.
Pinhole Aperture	2.0mm Diameter	Recessed pinhole for the camera to maintain a "stealth" appearance without visible glass lens.

2. Internal Skeleton: Thermal & Structural

To support the on-device intelligence engine and premium feel, the CRATE IQ uses a **CNC Aluminum 6061-T6 internal frame**.

Spec	Value	Technical Rationale
Thermal Mass	CNC Aluminum Frame	Acts as a direct heat sink for the i.MX 8M Nano processor (205 W/m·K conductivity). Pots make .
Mounting	Rigid Screw-Down	Components are screwed to the metal frame rather than snapped into plastic, ensuring long-term durability. C-A-Design .
Weighting	Bottom-Heavy	Strategic material placement to ensure the device feels balanced and "premium" in hand.

3. Haptic Intelligence

CRATE IQ replaces buzzes with "taps" using a **Linear Resonant Actuator (LRA)**.

Spec	Value	Technical Rationale
Actuator Type	10mm LRA (Z-Axis)	Faster response (<50ms) and crisper feedback compared to standard ERM motors. Boreas Financial content .
Feedback Logic	Single-Axis Tap	Directed output for UI scrolling and "Scan Success" signals, mimicking a mechanical feel. Piezo .
Driver	DRV2605L	Advanced haptic driver to support custom waveforms (e.g., "Matrix Scan Tick"). Youtube

4. Connectivity Technicalities

Component	Spec	TechnicalRationale
USB-C Port	16-pin SMT / Through-hole hybrid	Hybrid mounting for superior mechanical strength against port-tear Teenage
Antenna	None (Shielded)	Device is offline-first; internal layout is optimized for zero interference with the OCR imaging sensor.
Buttons	6mm Tactile Switch	High-cycle (1M+) mechanical switches with customized translucent caps for the "Soul" button.

5. Next Mechanical Steps

- ☐ **DFM Analysis:** Validate the "Smoke Amber" molding yields with Shenzhen CMs to ensure color consistency [Yajia](#)aluminum
- ☐ **Thermal Simulation:** Model heat dissipation during 5-minute continuous OCR session to ensure shell temperature stays below 35°C.
- ☐ **Drop Testing:** Simulate a 1.5m drop onto concrete to validate the "Aluminum-in-Polycarbonate" sandwich structural integrity